TABLE 11.4

Optimal MEWMA Control Charts [From Prabhu and Runger (1997)]

		$p = 4$		$p = 10$		$p = 20$	
δ	$ARL_0 =$	500	1000	500	1000	500	1000
0.5	λ	0.04	0.03	0.03	0.025	0.03	0.025
	H	13.37	14.68	22.69	24.70	37.09	39.63
	ARL_{min}	42.22	49.86	55.94	66.15	70.20	83.77
1.0	λ	0.105	0.09	0.085	0.075	0.075	0.065
	H	15.26	16.79	25.42	27.38	40.09	42.47
	ARL_{min}	14.60	16.52	19.29	21.74	24.51	27.65
1.5	λ	0.18	0.18	0.16	0.14	0.14	0.12
	H	16.03	17.71	26.58	28.46	41.54	43.80
	ARL_{min}	7.65	8.50	10.01	11.07	12.70	14.01
2.0	λ	0.28	0.26	0.24	0.22	0.20	0.18
	H	16.49	18.06	27.11	29.02	42.15	44.45
	ARL_{min}	4.82	5.30	6.25	6.84	7.88	8.60
3.0	λ	0.52	0.46	0.42	0.40	0.36	0.34
	H	16.84	18.37	27.55	29.45	42.80	45.08
	ARL_{min}	2.55	2.77	3.24	3.50	4.04	4.35

Note: ARL_0 and ARL_{min} are zero-state average run lengths.